CSE382 – Introduction to Machine Learning

End-to-End Machine Learning Project

City Hotel Booking Demand: Regression, Classification, and Clustering

Build a complete machine learning system using the **City Hotel (H2)** booking dataset. The project must include **regression**, **classification**, **unsupervised clustering**, dimensionality reduction, model comparison, **hyperparameter tuning**, saved inference artifacts, and a usable **CLI or GUI/web/desktop interface.**

Dataset	**Hotel Booking Demand – City Hotel (H2)**
Samples	**79,330 hotel bookings**
Variables	31 source variables
Required tasks	**Regression + Classification + Clustering**
Final submission	**End-to-end application; notebook-only submissions are not accepted**

1. Project Objective

The purpose of this project is to integrate the machine learning techniques studied in the course into one complete workflow. Students are expected to inspect and prepare the booking data, construct appropriate machine learning pipelines, compare multiple model families, tune hyperparameters, evaluate results correctly, and expose the final trained models through a usable software interface.

The dataset originates from hotel Property Management System (PMS) databases. The accompanying paper describes the published data as mixed raw and preprocessed data: some variables are direct booking fields while others were engineered from information stored in different database tables. The project is therefore focused on realistic tabular preprocessing rather than on receiving a fully model-ready feature matrix.

2. Dataset and Direct Download

Use the City Hotel dataset (H2) from the Hotel Booking Demand datasets by António, Almeida, and Nunes. The original publication describes H2 as 79,330 bookings with 31 variables. H2 contains bookings due to arrive between July 1, 2015 and August 31, 2017.

Direct H2.csv download: https://raw.githubusercontent.com/afaq-ahmad/Hotel-booking-demand-Case-Study/master/H2.csv

Original dataset article: https://pmc.ncbi.nlm.nih.gov/articles/PMC6297060/

The direct CSV link above is a public mirror of H2.csv. The original research article should be used as the authoritative source for variable definitions.

3. Files and Dataset Selection

The original dataset is distributed as two hotel datasets with the same 31-variable schema:

```
H1.csv  -> Resort Hotel  -> 40,060 bookings
H2.csv  -> City Hotel    -> 79,330 bookings
```

For this project, **use H2.csv only**. **Do not merge H1 and H2** for the required experiments. If you obtain a combined version from another source, isolate City Hotel observations before starting the project and verify that the resulting data matches the H2 schema.

4. Verified Source Labels and Meanings

The names below use the original H2 source labels exactly as documented for the dataset. If you choose to normalize names in code (for example, converting them to snake_case), document that transformation clearly and keep a one-to-one mapping to these source labels.

Exact source label	Type / role	Meaning
IsCanceled	Categorical / binary	1 if the booking was canceled; 0 otherwise.
LeadTime	Integer	Days between the booking entering the PMS and the arrival date.
ArrivalDateYear	Integer / temporal	Year of arrival.

ArrivalDateMonth	Categorical / temporal	Month of arrival.
ArrivalDateWeekNumber	Integer / temporal	Week number of the arrival date.
ArrivalDateDayOfMonth	Integer / temporal	Day of month of the arrival date.
StaysInWeekendNights	Integer	Number of weekend nights booked/stayed.
StaysInWeekNights	Integer	Number of weekday nights booked/stayed.
Adults	Integer	Number of adults.
Children	Integer	Number of children.
Babies	Integer	Number of babies.
Meal	Categorical	Meal package associated with the booking.
Country	Categorical	Country of origin code.
MarketSegment	Categorical	Market segment designation.
DistributionChannel	Categorical	Booking distribution channel.
IsRepeatedGuest	Categorical / binary	1 for a repeated guest; 0 otherwise.
PreviousCancellations	Integer	Number of previous bookings canceled by the customer.
PreviousBookingsNotCanceled	Integer	Number of previous bookings not canceled by the customer.
ReservedRoomType	Categorical	Code of the room type originally reserved.
AssignedRoomType	Categorical	Code of the room type ultimately assigned.
BookingChanges	Integer	Number of booking changes/amendments before check-in or cancellation.
DepositType	Categorical	Deposit arrangement associated with the booking.
Agent	Categorical / ID	Travel-agency identifier; may contain NULL.
Company	Categorical / ID	Company/entity identifier; may contain NULL.
DaysInWaitingList	Integer	Days the booking remained on the waiting list before confirmation.
CustomerType	Categorical	Type of booking/customer.
ADR	Numeric	Average Daily Rate.
RequiredCarParkingSpaces	Integer	Number of requested car-parking spaces.
TotalOfSpecialRequests	Integer	Number of special requests made by the customer.
ReservationStatus	Categorical	Final reservation status.
ReservationStatusDate	Date	Date on which the last reservation status was set.

5. General Technical Requirements

- **Python must be used** for the machine learning component.
- Libraries such as scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, Optuna, pandas, NumPy, and Matplotlib are permitted.
- From-scratch implementation of machine learning algorithms is **not required**.
- Notebook files may be used for exploration but **must not be the main submitted solution**.
- The final project must be modular and runnable through a **CLI or a GUI/web/desktop application**.
- All important preprocessing and modeling steps must be reproducible.
- Use fixed random seeds where applicable.
- Store trained preprocessing objects, encoders, dimensionality-reduction objects, and final models as reusable artifacts.
- Inference must load saved artifacts rather than retrain the model every time the application starts.

Team and GitHub Requirements

Team registration: Link

- Each project team consist of **exactly 4 to 5 students**.
- Write your **ID ONLY** and your Name will appear Automatically
- There is a Sheet named **"Finding Team"**, you can write your ID there if you can't find a Team and you will be distributed in a random team
- It is allowed that Teams members be from different sections or different Majors, but handling discussion time will be your responsibility
- All project work must be maintained in a **GitHub repository from the beginning of the project**.
- Each team member must contribute using **their own GitHub account**. The **commit history will be reviewed** to evaluate individual participation and the development process.
- Commits should represent **meaningful, incremental work** throughout the project. Uploading the complete project only near the deadline **does not demonstrate individual contribution**.
- **Project grades are individual**. Team members are **not guaranteed to receive the same grade**; individual discussion performance, technical understanding, and verifiable **GitHub contribution** may affect each student's final project grade.

6. Data Understanding and Preprocessing

- Inspect dimensions, variable types, duplicates, missing values, invalid or unusual values, category cardinality, and data ranges.
- Produce suitable exploratory plots and summary statistics.
- Investigate skewed variables and outliers.
- Handle missing values using a justified strategy.
- Encode categorical variables appropriately.
- Scale or normalize variables where required by the selected model.
- Derive useful features only when they are justified and reproducible.
- Create a clear train/validation/test strategy or a train/test strategy combined with cross-validation.
- Use pipelines or an equivalent structured mechanism so preprocessing is applied consistently.
- Audit the complete workflow for **direct and indirect data leakage**.

7. Task A – Regression

Regression target: ADR. Build a model that predicts the Average Daily Rate associated with a booking.

Students must determine which source variables are appropriate predictors for this task and justify the assumed prediction point. Do not assume that every field in the CSV is automatically a valid input simply because it is present.

Regression model requirement:

Select and evaluate **at least three different regression models**. Each selected model must be evaluated in both an **untuned/baseline configuration** and a **hyperparameter-tuned configuration**.

Candidate models include:

- Linear Regression (baseline).
- K-Nearest Neighbors Regressor.
- Decision Tree Regressor.
- Random Forest Regressor.
- XGBoost Regressor or another approved gradient-boosted tree implementation.
- Support Vector Regression, including an RBF-kernel experiment.
- Multilayer Perceptron / Neural Network Regressor.

Required regression metrics:

- MAE.
- MSE and/or RMSE.
- R^2.
- Cross-validation or validation-set results used for model selection.
- Final test-set results.

The regression pipeline must avoid direct and indirect target leakage. Consider the timing, meaning, and relationship of each candidate feature to ADR, and document your decisions.

8. Task B – Classification

Classification target: IsCanceled. Build a binary classifier that predicts whether a booking will eventually be canceled.

Define a reasonable point in the booking lifecycle at which this prediction is assumed to be made. Your feature selection must be consistent with that prediction point.

Classification model requirement:

Select and evaluate **at least three different classification models**. Each selected model must be evaluated in both an **untuned/baseline configuration** and a **hyperparameter-tuned configuration**.

Candidate models include:

- Gaussian Naive Bayes or another justified Bayesian classifier.
- Logistic Regression / linear classifier.
- K-Nearest Neighbors.
- Decision Tree.
- Random Forest.
- XGBoost classifier or another approved gradient-boosted tree implementation.
- Support Vector Machine, including at least one kernel-based experiment.
- Multilayer Perceptron / Neural Network classifier.

Required classification metrics:

- Accuracy.
- Precision.

- Recall.
- F1-score.
- Confusion matrix.
- ROC-AUC and/or PR-AUC where appropriate.
- Cross-validation results.
- Final test-set performance.

The classification workflow must avoid both direct and indirect leakage. Inspect the semantic meaning and timing of every candidate feature, especially variables whose values may change as a booking progresses through its lifecycle. The specification intentionally does not provide a list of columns to remove.

9. PCA and LDA

- Apply PCA to an appropriately prepared numerical representation of the input features.
- Report explained variance and justify the number of retained principal components.
- Compare at least two supervised models with and without PCA.
- Apply LDA to the classification task where appropriate.
- Compare original features, PCA features, and LDA features for at least one classification model.
- Integrate all fitted transformations correctly into the modeling pipeline.

10. Task C – Clustering

Use the H2 records to discover natural booking/customer segments. The clustering task must remain genuinely unsupervised: clusters should be discovered from selected booking attributes rather than being supplied with an outcome.

Clustering model requirement:

Select and evaluate **at least three different clustering algorithms**. Each selected algorithm must be compared in an **initial/baseline configuration** and after systematic **hyperparameter/parameter optimization** using appropriate unsupervised criteria.

Candidate algorithms include:

- K-Means.
- Agglomerative Hierarchical Clustering.
- DBSCAN.
- Gaussian Mixture Model (GMM) or another justified clustering algorithm.

Required clustering analysis:

- Choose and justify the subset of source variables used for clustering.
- Handle categorical variables appropriately.
- Scale or transform variables when required by the distance/proximity measure.
- Investigate a sensible number of clusters rather than selecting it arbitrarily.
- Use internal evaluation measures such as Silhouette Score and Davies–Bouldin Index.

- Produce a PCA-based or other justified two-dimensional visualization of cluster structure.
- Produce a dendrogram for hierarchical clustering where computationally practical.
- Interpret the discovered clusters using meaningful booking characteristics.

Because H2 contains 79,330 rows, full agglomerative clustering may be computationally expensive. A justified sampling or representative-subset strategy is allowed for hierarchical clustering, provided the method is documented. K-Means should use the full dataset where practical.

The clustering workflow must also be audited for leakage. Think carefully about whether a chosen feature encodes a final outcome, future event, or information that would make the discovered grouping artificially dependent on an answer that should not have been supplied.

11. Hyperparameter Optimization

Hyperparameter optimization is mandatory. GridSearchCV, RandomizedSearchCV, Optuna, or another approved optimizer may be used.

- For regression, tune all selected models; at least three regression models must therefore have both baseline and tuned results.
- For classification, tune all selected models; at least three classification models must therefore have both baseline and tuned results.
- Among the selected supervised models, include at least one ensemble/boosting model such as Random Forest or XGBoost.
- For clustering, optimize the parameters of all selected algorithms (at least three) using appropriate unsupervised criteria. Do not use the supervised target labels to choose clustering hyperparameters.
- Document the search space and optimization metric.
- Do not use the final test set for repeated hyperparameter selection.
- Store and report the best configuration found.

12. Model Comparison and Selection

- Use a consistent experimental protocol so comparisons are fair.
- Record preprocessing configuration, feature representation, hyperparameters, validation/CV results, test results, and training time.
- For **every selected model/algorithm** in each task, report both the **baseline (untuned)** and **tuned/optimized** results.
- Select the deployed regression and classification models using a clearly stated model-selection criterion.
- Do not report only the single best model; comparison across model families is required.

13. End-to-End Application Requirement

The final system **must be usable without opening source code or a notebook**. Students may implement a command-line interface or a graphical interface.

- CLI application.

- Streamlit or Gradio application.
- Flask or FastAPI backend with a web frontend.
- React frontend connected to a Python API.
- PyQt or Tkinter desktop application.

At minimum, the final application must allow the user to:

- View basic dataset information and analysis results.
- Train models or load previously trained artifacts.
- View model comparison results.
- Run or inspect hyperparameter tuning results.
- Enter/select booking information and obtain an ADR regression prediction.
- Enter/select booking information and obtain an IsCanceled classification prediction.
- Run or view the clustering analysis.
- Display the selected model and relevant evaluation information.

14. Suggested CLI Commands (Example Only)

```
python main.py analyze
python main.py train --task regression --model random_forest
python main.py train --task classification --model xgboost
python main.py tune --task regression --model xgboost
python main.py evaluate --task classification --model all
python main.py cluster --algorithm kmeans
python main.py predict --task regression
python main.py predict --task classification
```

15. Recommended Repository Structure

```
project/
├── README.md
├── requirements.txt
├── config.yaml
├── main.py
├── data/
├── src/
│   ├── data/
│   ├── preprocessing/
│   ├── features/
│   ├── regression/
│   ├── classification/
│   ├── clustering/
│   ├── tuning/
│   ├── evaluation/
│   └── inference/
├── artifacts/
├── outputs/
└── tests/
```

An equivalent clean and modular architecture is acceptable.

16. Leakage Audit Requirement

Each team must include a dedicated **Leakage Audit** section in the report. The objective is to show that the team understands how leakage can appear indirectly, not simply that it knows the term.

- Identify possible leakage risks in the ADR regression task.
- Identify possible leakage risks in the IsCanceled classification task.
- Identify possible leakage risks in the clustering task.
- Explain why each selected feature is appropriate at the stated prediction/analysis point.
- Explain how preprocessing, feature engineering, dimensionality reduction, tuning, and final evaluation are separated.

The specification intentionally does not provide a complete list of leakage-prone variables. Recognizing those risks is part of the assessed work.

17. Deliverables

1. Complete source-code repository submitted as a ZIP file, with the GitHub repository link included in the README so the full commit history can be reviewed.
2. README containing environment setup and exact run commands.
3. requirements.txt or equivalent dependency specification.
4. Saved preprocessing/model artifacts required for inference must be generated and available locally for the live discussion; exclude trained model and fitted pipeline files from the submitted ZIP to keep its size manageable.
5. Evaluation tables and figures.
6. A project report documenting preprocessing, model experiments, baseline-vs-tuned comparisons, hyperparameter optimization, leakage audit, clustering interpretation, and final model selection.
7. A live demonstration of the final end-to-end system.

Notebook files may be included as optional supplementary exploration only. They are **not accepted as the main implementation** or as a replacement for the final application.

18. Project Milestones, Submission, and Discussion Schedule

The project is divided into two milestones. All teaching weeks start on Friday. Week 3 runs from 2–8 October 2026; 3 October falls within Week 3. Week 8 is the midterm examination week, starting 6 November 2026. Week 15, the final teaching week, starts 25 December 2026. No milestone deadline or project discussion is scheduled during Week 8.

Stage	Deadline / period	Purpose
Milestone 1	Thursday, 5 November 2026 — end of Week 7	Data preparation and initial baseline experiments; submission only.
Midterms	Week 8: 6–12 November 2026	Midterm examinations.
Milestone 2	Friday, 11 December 2026 — Week 13	Complete final project submission.
Final discussions	Week 14: Friday, 18–Thursday, 24 December 2026	Live demonstration, evaluation, and individual discussion.

Milestone 1 — Data Preparation and Baseline Experiments

- Set up the team's GitHub repository, identify all members and their accounts, establish the modular code structure, and provide environment setup and run instructions.
- Load and verify H2.csv; complete data understanding and EDA, including missing values, duplicates, variable types, unusual values, distributions, and suitable plots.
- Define the prediction point, task-specific feature choices, and leakage risks for regression, classification, and clustering. Document a reproducible train/validation/test or cross-validation strategy.
- Implement reusable preprocessing pipelines, including justified missing-value handling, categorical encoding, scaling, and any feature engineering. Fit data-dependent supervised preprocessing on training data only.
- Run initial untuned experiments with at least one regression model, one classification model, and one clustering algorithm. Report appropriate preliminary metrics and a brief interpretation; keep the final supervised test set untouched for final evaluation.
- Submit a progress report containing completed work, initial results, a preliminary leakage audit, member contributions, and the plan for the remaining models, tuning, PCA/LDA, and final application.

Discussion policy: There is no scheduled or mandatory discussion for Milestone 1. Any team that wishes to discuss its progress may request an optional discussion with its dedicated TA. The final project discussion is held for Milestone 2.

Milestone 2 — Complete End-to-End Project

- Complete all requirements in this specification, incorporating any necessary corrections to Milestone 1 work.
- Evaluate at least three regression models, three classification models, and three clustering algorithms. Include both baseline and tuned/optimized results for every selected model or algorithm, the search spaces, optimization criteria, and best configurations.
- Complete PCA and LDA experiments and the required comparisons; provide final model comparison tables, justified model selection, final supervised test-set evaluation, and clustering metrics, visualizations, and interpretation.
- Deliver the runnable CLI or GUI/web/desktop application, including regression and classification inference using saved preprocessing/model pipelines and access to the required analysis and comparison results.
- Submit the final report, complete leakage audit, figures/tables, dependency specification, exact reproduction commands, GitHub repository link, and identifiable contributions for all team members.
- Prepare the trained models and fitted preprocessing/dimensionality-reduction artifacts locally for the discussion. Verify beforehand that the application loads these artifacts and runs successfully.

Submission Format and Live Evaluation

- For each milestone, submit one ZIP file of the repository at that milestone's submitted version. Include source code, configuration, README, dependencies, the applicable report, and result tables/figures. Include the GitHub repository URL and the submitted commit SHA in the README; the online commit history will be checked.
- Do not include trained model files or serialized fitted preprocessing/model pipelines in the ZIP, to keep submission size manageable. Include the code, configurations, and exact commands needed to regenerate them. Saved artifacts remain mandatory for final inference and must be available locally at the discussion.

- Evaluation will be live during the Milestone 2 discussion. The team must run the submitted application, demonstrate predictions and required analyses, and explain its experimental results. Each member may be asked about any part of the workflow; individual understanding and GitHub contributions will be assessed.
- Milestone 2 is due on 11 December 2026, one week before the discussion week starts on 18 December. Discussions take place during 18–24 December, with the dedicated TA assigning each team's slot. All discussions must finish before Week 15 starts on 25 December 2026.
- The existing grading weights remain unchanged: Discussion 40%, Report 20%, and Technical 40%. The two milestones organize delivery; they do not introduce additional grading weights.

19. Minimum Expected Comparison

Area	Minimum expected work
Regression	At least 3 regression models; each shown before and after hyperparameter tuning; consistent evaluation metrics and final comparison.
Classification	At least 3 classification models; each shown before and after hyperparameter tuning; leakage-aware IsCanceled prediction and consistent final comparison.
PCA / LDA	Original feature space vs PCA; original/PCA/LDA comparison for at least one classifier.
Clustering	At least 3 clustering algorithms; baseline and optimized parameter settings; cluster-count/parameter justification, metrics, visualization, and interpretation.
Optimization	Baseline vs tuned/optimized results for every selected model/algorithm across Regression, Classification, and Clustering.
Deployment	Saved trained pipelines used for new-sample regression and classification inference.

20. Grading Criteria (100%)

Grading Component	Weight
Discussion	**40%**
Report	**20%**
Technical	**40%**

Discussion (40%): **assessed individually** during the project discussion/demo. Students must demonstrate understanding of the complete workflow, justify preprocessing and feature choices, explain model behavior and results, discuss leakage risks, and answer questions about any part of the implementation. Individual discussion performance may result in **different grades** for members of the same team.

Report (20%): assessed on clarity, completeness, experimental documentation, figures/tables, interpretation of results, leakage audit, model comparison, and reproducibility instructions.

Technical (40%): **all implementation and machine-learning technical requirements** are assessed inside this component. The 40% technical grade is distributed as follows:

Technical Component	Weight (of total project grade)
Data understanding, EDA, cleaning, and preprocessing	**5%**
Regression pipeline, 3+ models, tuning, and evaluation	**6%**
Classification pipeline, 3+ models, tuning, and evaluation	**6%**
Clustering, 3+ algorithms, optimization, and interpretation	**5%**
PCA and LDA experiments	**4%**
Hyperparameter optimization and model comparison	**5%**
Leakage prevention and experimental integrity	**4%**
End-to-end application and saved inference pipeline	**3%**
Code quality and reproducibility	**2%**

21. Important Rules

- **Use H2.csv (City Hotel) for the required experiments.**
- Do not merge H1 and H2 for the main project experiments.
- Use the source labels in Section 4 when referring to the original columns; if you rename them in code, document the mapping.
- **Do not submit a notebook-only solution.**
- Package implementations of the required ML algorithms are allowed and expected.
- **For each of the three tasks (Regression, Classification, and Clustering), evaluate at least three different models/algorithms and compare baseline (untuned) performance against tuned/optimized performance.**
- Do not assume every dataset column is safe to use as a predictor.
- **Do not repeatedly inspect final test performance while choosing preprocessing steps or hyperparameters.**
- Do not report only the best-performing model.
- Do not hard-code predictions; inference must use the saved preprocessing and model pipeline.
- **Every team member must understand the complete workflow and may be asked about any component during the demonstration.**
- Each team contain **exactly 4-5 students**.
- **GitHub is mandatory**. The repository and **commit history will be checked** as part of the assessment.
- Every student must have identifiable, **meaningful contributions** in the repository. **Individual grades may differ** between members of the same team.

22. Dataset Reference

Nuno António, Ana de Almeida, and Luís Nunes. “Hotel booking demand datasets.” Data in Brief, Volume 22, 2019, pp. 41–49.

Original article: https://pmc.ncbi.nlm.nih.gov/articles/PMC6297060/

Direct H2.csv mirror: https://raw.githubusercontent.com/afaq-ahmad/Hotel-booking-demand-Case-Study/master/H2.csv

